

Logarithmic Functions: Problems and Solutions

APMA Faculty
University of Virginia

August 9, 2025

Problem 1. *Simplify the expression using logarithmic properties:*

$$\log_5(25) + \log_5(5)$$

Solution:

$$\log_5(25) + \log_5(5) = \log_5(25 \times 5) = \log_5(125) = \log_5(5^3) = 3$$

Problem 2. *Solve for x using the properties of logarithms:*

$$\log_2(x) + \log_2(8) = 4$$

Solution:

$$\log_2(x \times 8) = 4 \quad \Rightarrow \quad \log_2(8x) = 4 \quad \Rightarrow \quad 8x = 2^4 = 16 \quad \Rightarrow \quad x = 2$$

Problem 3. *Simplify:*

$$\log_3(81) - \log_3(9)$$

Solution:

$$\log_3\left(\frac{81}{9}\right) = \log_3(9) = \log_3(3^2) = 2$$

Problem 4. Use the change of base formula to evaluate the logarithm:

$$\log_4(16)$$

Solution:

$$\log_4(16) = \frac{\log_{10}(16)}{\log_{10}(4)} = \frac{\log_{10}(2^4)}{\log_{10}(2^2)} = \frac{4 \log_{10}(2)}{2 \log_{10}(2)} = 2$$

Problem 5. Simplify the expression:

$$\frac{1}{3} \log_2(8) + \frac{1}{2} \log_2(4)$$

Solution:

$$\frac{1}{3} \log_2(8) + \frac{1}{2} \log_2(4) = \frac{1}{3} \times 3 + \frac{1}{2} \times 2 = 1 + 1 = 2$$

Problem 6. Solve for y in the equation:

$$\log_{10}(y) = \log_{10}(5) + \log_{10}(2)$$

Solution:

$$\log_{10}(y) = \log_{10}(5 \times 2) = \log_{10}(10) = 1 \quad \Rightarrow \quad y = 10$$

Problem 7. Express the logarithmic expression as a sum or difference of logarithms:

$$\log_b \left(\frac{x^2 \cdot y^3}{z} \right)$$

Solution:

$$\log_b \left(\frac{x^2 \cdot y^3}{z} \right) = \log_b(x^2) + \log_b(y^3) - \log_b(z) = 2 \log_b(x) + 3 \log_b(y) - \log_b(z)$$

Problem 8. *Use the properties of logarithms to simplify:*

$$\log_7(7^4) - 2 \log_7(7)$$

Solution:

$$\log_7(7^4) - 2 \log_7(7) = 4 \log_7(7) - 2 \log_7(7) = 4 \times 1 - 2 \times 1 = 4 - 2 = 2$$

Problem 9. *Solve the logarithmic equation for x :*

$$2 \log_3(x) = \log_3(9)$$

Solution:

$$2 \log_3(x) = \log_3(9) \quad \Rightarrow \quad \log_3(x^2) = \log_3(9) \quad \Rightarrow \quad x^2 = 9 \quad \Rightarrow \quad x = \pm 3$$

But $\log_3(x)$ is undefined for $x < 0 \Rightarrow x = 3$ [Domain restriction]

Problem 10. *Express the following logarithmic expression using natural log:*

$$\log_5(50)$$

Solution:

$$\log_5(50) = \frac{\ln(50)}{\ln(5)}$$

Problem 11. *Simplify* $10^{-\log_{10}(2x)}$

$$10^{-\log_{10}(2x)} = \frac{1}{10^{\log_{10}(2x)}} = \frac{1}{2x}$$

Problem 12. *Simplify* $(2^{\log_2(x)})^{-1}$

$$(2^{\log_2(x)})^{-1} = x^{-1} = \frac{1}{x}$$

Problem 13. *Simplify* $e^{-3\ln(2)}$

$$e^{-3\ln(2)} = e^{\ln(2^{-3})} = 2^{-3} = \frac{1}{8}$$

Problem 14. *Simplify* $e^{-\frac{1}{2}\ln(16)}$

$$e^{-\frac{1}{2}\ln(16)} = e^{\ln(16^{-1/2})} = 16^{-1/2} = \frac{1}{\sqrt{16}} = \frac{1}{4}$$

Problem 15. *Solve the logarithmic equation for x :*

$$\log_3(x) + \log_3(x - 2) = 1$$

Solution:

$$\log_3(x) + \log_3(x - 2) = 1 \quad \Rightarrow \quad \log_3(x(x - 2)) = 1$$

$$\Rightarrow \log_3(x^2 - 2x) = 1$$

$$\Rightarrow x^2 - 2x = 3^1 = 3$$

$$\Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0 \Rightarrow x = 3 \text{ or } x = -1$$

Check domain: $\log_3(x)$ and $\log_3(x - 2)$ are undefined for $x \leq 0$ or $x \leq 2$

$$\Rightarrow \text{Valid solution: } \boxed{x = 3}$$

Problem 16. Solve the logarithmic equation for x :

$$\ln(x) - \ln(x - 3) = \ln(2)$$

Solution:

$$\ln(x) - \ln(x - 3) = \ln(2) \Rightarrow \ln\left(\frac{x}{x - 3}\right) = \ln(2)$$

$$\Rightarrow \frac{x}{x - 3} = 2$$

$$\Rightarrow x = 2(x - 3) = 2x - 6 \Rightarrow x - 2x = -6 \Rightarrow -x = -6 \Rightarrow x = 6$$

Check domain: $x > 0$ and $x - 3 > 0 \Rightarrow x > 3$

$$\Rightarrow \text{Valid solution: } \boxed{x = 6}$$